Pedro Saraiva

ML ENGINEER

ABOUT ME

ML Engineer working across the whole model lifecycle: data pipelines, training, evaluation and production deployment. I have shipped computer vision for robotics, NLP and LLM systems for legal and government use, and MLOps infrastructure. Most of my day-to-day work is figuring out why a model behaves the way it does (unstable training, misleading metrics, a pipeline that leaks data) and fixing it.

COMPETENCES

ML stack: Python (advanced), PyTorch, Hugging Face, scikit-learn, TensorFlow, OpenCV, LangChain.

Training & evaluation: experiment design, validation splits, metric selection, debugging training failures, anomaly detection, time series.

MLOps: ML pipelines, model monitoring, CI/CD for ML, Docker, Git, cloud deployment.

Domains: Computer Vision, NLP / LLMs, Robotics, Legal Analytics.

Engineering: SQL, data engineering, system architecture, production-grade code.

LANGUAGES

Portuguese: Native

English: Professional working proficiency

CONTACT

pedro.a.saraiva08@gmail.com

linkedin.com/in/pedro-a-maciel-saraiva

WORK EXPERIENCE

AI / MLOps Engineer **2026 – present**

Confi

- Build and operate ML and LLM-agent systems in production, covering deployment, monitoring and pipeline automation.

AI Engineering Intern

Encore Asset Management

- Built Janus, a multi-agent LLM system where new data sources can be plugged in to generate equity research reports.
- Analysts use these reports to make buy and sell decisions on companies listed on B3, the Brazilian stock exchange.

LLM Engineer — Brazil's IRS (Receita Federal) **2025 – 2026**

CEIA — Center of Excellence in Artificial Intelligence, UFG

- Built a locally hosted LLM system that helps citizens with tax regularization.

NLP Engineer — Educational Chatbots **2024 – 2026**

CEIA — Center of Excellence in Artificial Intelligence, UFG

- Designed and deployed NLP models for an educational chatbot supporting robotics learning and public-service exam prep.
- Owned the end-to-end cycle, from data collection and training to inference and response-quality optimization.

NLP Engineer — Jurimetrics **2024 – 2025**

CEIA — Center of Excellence in Artificial Intelligence, UFG

- Built analytics pipelines that extract insights from legal data to support lawyers' decisions.
- Developed NLP classifiers that categorize and analyze diverse legal case types.

Aerial Robotics Researcher **2023 – 2026**

Pequi Mecânico — Robotics Research Group, UFG

- Developed computer vision models for autonomous drone perception and navigation.
- Integrated software and hardware for real-time inference on edge devices.

SELECTED PROJECTS

- **Credit Card Fraud Detection:** autoencoder-based anomaly detection on highly imbalanced data.
- **Electricity Price Prediction:** time series forecasting models.
- **Crop Monitoring with EMBRAPA:** quadrupedal and aerial robots for monitoring genetically modified soy.
- **OreIA (co-founder):** IoT startup for smart audio control in shopping malls.

EDUCATION

B.Sc. in Artificial Intelligence **2023 – 2026**

Federal University of Goiás (UFG)